

LAB 9: Wireless Network Setup and Packet Analysis using Cisco Packet Tracer

OBJECTIVES

- To set up and configure a wireless router in Cisco Packet Tracer and establish a functional wireless local area network (WLAN).
- To connect wireless client devices to the network and enable automatic IP address allocation through the DHCP service.
- To validate secure wireless communication by applying WPA2 encryption and confirming end-to-end connectivity between clients.

THEORY

1. Wireless Networks and WLANs

Wireless networks transmit data using radio frequency signals instead of physical cables. This provides flexibility and mobility for devices within the network. A Wireless Local Area Network (WLAN) is created when a wireless router or access point broadcasts signals that nearby devices can connect to. In this lab, a wireless router acts as the access point, providing wireless connectivity and assigning IP addresses to laptop clients through DHCP.

2. SSID and Network Identification

An SSID (Service Set Identifier) is the name used to identify a wireless network. The router broadcasts this name so nearby devices can detect and connect to the network. Users select the correct SSID from the list of available networks and enter the required credentials. In this lab, the SSID is configured as HCOE, which clients select to connect to the wireless network.

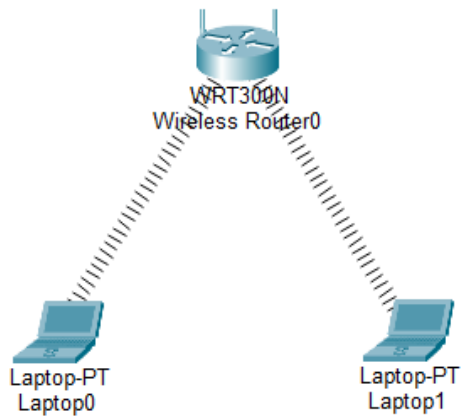
3. Wireless Security -WPA2 and AES Encryption

Wireless networks are more vulnerable to unauthorized access because signals travel through open air. WPA2 (Wi-Fi Protected Access 2) improves security by encrypting transmitted data. When combined with AES encryption, it provides strong protection for wireless communication. In WPA2-Personal mode, users must enter a shared password to connect, making it suitable for small or medium networks like the one used in this lab.

4. DHCP in Wireless Networks

After connecting to the wireless network, devices require an IP address to communicate. The router's DHCP server automatically assigns IP addresses, subnet masks, and default gateways to connected clients. This automatic configuration simplifies network management and allows devices to join the network without manual IP setup.

NETWORK TOPOLOGY



Device	IP Assignment	SSID	Security
WirelessRouter0	192.168.0.1 (Static)	HCOE	WPA2-AES
Laptop0	DHCP	HCOE	WPA2-AES
Laptop1	DHCP	HCOE	WPA2-AES

CONFIGURATION

Router Configuration Summary

Parameter	Configured Value
IP Address	192.168.0.1
Subnet Mask	255.255.255.0
DHCP Server	Enabled
SSID	HCOE
Network Mode	Mixed
Channel	Auto
Security Mode	WPA2-Personal
Encryption	AES
Passphrase	hcoe@123

Procedure

Step 1: Basic Router Setup

1. Click on WirelessRouter0 in the Packet Tracer workspace.
2. Open the GUI tab and navigate to Setup.
3. Enter the following values:
 - IP Address: 192.168.0.1
 - Subnet Mask: 255.255.255.0
 - DHCP Server: Enabled
4. Save the settings before proceeding.

Step 2: Wireless Configuration

1. In the router GUI, navigate to Wireless → Basic Wireless Settings.
2. Configure the following parameters:
 - Network Mode: Mixed
 - SSID: HCOE
 - Channel: Auto
3. Click Save Settings to apply the wireless configuration.

Step 3: Wireless Security

1. Go to Wireless → Wireless Security in the router GUI.
2. Set the security parameters as follows:
 - Security Mode: WPA2-Personal
 - Encryption: AES
 - Passphrase: hcoe@123
3. Save the configuration.

Step 4: Connecting Laptops to the Wireless Network

1. On each laptop, open the Physical tab and replace the default wired NIC with a WPC300N wireless adapter.
2. Power the laptop back on, then go to Desktop → PC Wireless.
3. Under the Connect tab, click Refresh and select the HCOE network from the scan results.

4. Enter the passphrase hcoe@123 when prompted and click Connect.
5. Confirm that the adapter shows an active link and that a DHCP address has been assigned.

OBSERVATIONS



Wireless Network Detection

Laptop1 detected the wireless network HCOE broadcast by WirelessRouter0 with strong signal strength. The network operated in Infrastructure mode with mixed B/G/N support and WPA2-PSK security, confirming the router's wireless configuration.

Wireless Authentication and DHCP Assignment

Both laptops successfully connected to the HCOE network using the WPA2 passphrase. After authentication, each device automatically received an IP address from the router's DHCP server, confirming successful association and network configuration.

Packet Analysis (Simulation Mode)

In Simulation mode, the DHCP process was observed through the Discover, Offer, Request, and Acknowledge sequence. ARP packets were also generated to resolve IP-to-MAC addresses before data transmission.

Connectivity Verification

A ping test between Laptop0 and Laptop1 was successful with 0% packet loss, confirming proper wireless communication between the devices.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>
ping 192.168.0.101

Pinging 192.168.0.101 with 32 bytes of data:

Reply from 192.168.0.101: bytes=32 time=38ms TTL=128
Reply from 192.168.0.101: bytes=32 time=24ms TTL=128
Reply from 192.168.0.101: bytes=32 time=23ms TTL=128
Reply from 192.168.0.101: bytes=32 time=30ms TTL=128

Ping statistics for 192.168.0.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 23ms, Maximum = 38ms, Average = 28ms
```

The wireless network was successfully configured in Cisco Packet Tracer. The router broadcast the SSID HCOE with WPA2-AES security and DHCP enabled. Both laptops connected to the network, received IP addresses automatically, and communicated successfully through ping tests.

DISCUSSION AND CONCLUSION

This lab demonstrated the setup and security of a wireless network. Configuring the SSID and WPA2 passphrase-controlled network access, while DHCP automated IP address assignment. Simulation mode helped visualize DHCP and ARP packet exchanges, showing the steps required before data communication can occur. The exercise highlighted the importance of proper authentication, IP configuration, and address resolution in wireless networking.

The lab objectives were successfully achieved by configuring a wireless router with SSID, WPA2 security, and DHCP service. Both laptops connected wirelessly, obtained IP addresses automatically, and communicated successfully. The lab reinforced the concepts of WLAN setup, wireless security, and network communication.